

A Project on
ELDERLY CARE MONITORING SYSTEM
(Internet of Things Laboratory- ECE 305)

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AKNOWLEDGEMENT

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We are also truly thankful to the **university library staff** for their cooperation and timely assistance. Their help in providing access to important online research materials, reference books, and digital databases greatly contributed to the quality and depth of our project. Their dedication and support made our research process smoother and more effective.

We have made every possible effort to acknowledge, credit, and appreciate all the sources of information, ideas, and inspiration that supported us during the development of this project. We have followed academic integrity to the best of our knowledge while preparing this work.

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ABSTRACT.

We would like to convey our heartfelt gratitude to **Prof. Dr. Rupesh Kumar**, our respected professor, for providing us with the valuable opportunity to work on the project titled “**Elderly Care Monitoring System.**” His constant support, thoughtful suggestions, and encouragement played a key role in the successful completion of this work.

We also sincerely thank the dedicated staff of the **university library** for enabling our access to essential online resources, research materials, and databases that greatly contributed to our project. Their timely help and cooperation made our research process smooth and effective.

We are also grateful to all our classmates and well-wishers who motivated us and offered useful feedback during different stages of the project.

We have made every effort to acknowledge and appropriately credit all the references, ideas, and content we have used, to the best of our understanding and knowledge.

INTRODUCTION:

The Elderly Care Monitoring System is an IoT-based safety solution developed to continuously monitor the health and safety of senior citizens. The system uses an ESP32 microcontroller as the central unit, enabling wireless connectivity and live monitoring. An MPU6050 accelerometer and gyroscope sensor detects sudden movements or abnormal acceleration patterns that may indicate a fall. When a fall is detected, the system immediately activates a buzzer alert to notify nearby caregivers. A button is also provided so the elderly person can stop the alarm if the fall is not serious. Accurate location tracking is enabled through a GPS module, which provides real-time latitude, longitude, and satellite information. All sensor data is displayed on a web dashboard hosted by the ESP32, allowing caregivers to monitor the user remotely. The dashboard updates automatically and reflects the current status of fall detection, GPS position, and system activity. A medicine reminder feature is included to alert the user at regular intervals to take their medication on time. This makes the system helpful not only in emergencies but also in daily health management. The integration of GPS tracking and fall detection ensures quick response during critical situations. The project aims to provide a reliable, affordable, and efficient IoT solution for elderly safety and independence. Overall, the system enhances care quality by offering continuous monitoring and timely alerts for better elderly support. The overall objective of the project is to achieve efficient, accurate, and real-time fall detection on resource constrained hardware while enabling instant communication based alerts for timely intervention.

METHODOLOGY:

The proposed system was developed through structured stages, from sensor setup to real-time monitoring and alert generation. The key steps are:

1. **Sensor Setup and Initialization:**

The ESP32 initializes the MPU6050 motion sensor and GPS module, preparing them to continuously capture movement and location data.

2. **Continuous Data Collection:**

The MPU6050 records acceleration and gyroscope values in real time, while the GPS updates latitude, longitude, and satellite strength. This data forms the basis for fall detection and tracking.

3. **Fall Detection Process:**

Acceleration values are constantly checked against a predefined threshold. Any sudden spike or abnormal movement is treated as a potential fall.

4. **Emergency Alert Activation:**

When a fall is detected, the system triggers the buzzer to alert nearby individuals. A push-button allows the elderly user to cancel the alert if they are safe, avoiding false alarms.

5. **Real-Time GPS Tracking:**

The GPS module continuously updates the user's location, helping caregivers quickly locate them during emergencies.

6. **Medicine Reminder Function:**

A timed reminder activates alerts at regular intervals to ensure that the user does not miss their scheduled medication.

7. **Web Dashboard Monitoring:**

The ESP32 hosts a real-time web dashboard displaying fall status, GPS location, satellite count, buzzer activity, and medicine alerts. Caregivers can check the user's condition anytime.

FLOWCHART:

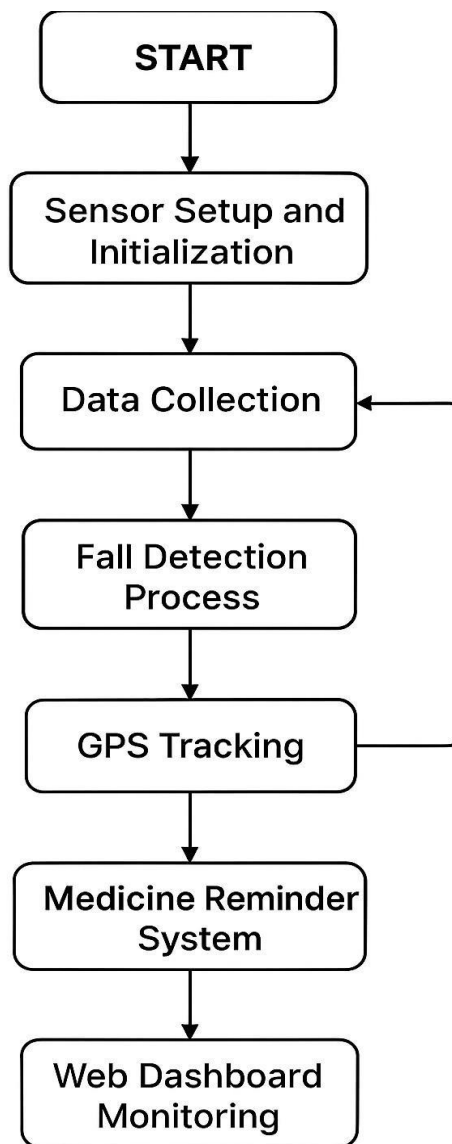


Fig 1:

The system starts by setting up the sensors. It then collects movement and location data continuously. The fall detection block checks for any sudden or abnormal movement. GPS tracking updates the user's location. The medicine reminder gives alerts at the right time. Finally, all information is shown on a web dashboard for caregivers to monitor easily.

WEB DASHBOARD INTEGRATION (ESP32) :

The proposed system integrates a real-time web dashboard hosted directly on the ESP32 using its Wi-Fi capability. Once the device is powered on and connected to the network, caregivers can access the dashboard through any browser without requiring additional applications. The dashboard continuously displays essential information such as fall status, GPS coordinates, satellite count, buzzer activity, and medicine alert notifications. The page refreshes automatically to ensure that data is always up to date. This interface allows caregivers to monitor the user remotely and respond immediately to emergency events. It eliminates the need for external servers, making the system more cost-effective and user friendly.

Why we use ☐ Web Dashboard....

Real-Time Monitoring:

The web dashboard allows us to see live updates from the system at any moment, which is very important for safety and health applications.

Instant Data Display:

As soon as the ESP32 collects data, it is immediately shown on the dashboard, helping caregivers respond quickly during emergencies.

No Extra Cost:

The dashboard works using normal Wi-Fi, so there are no additional charges for sending alerts or updates.

Shows Detailed Information:

It can display multiple types of data such as GPS location, fall detection status, satellite count, and system condition on one screen in a clear way.

Accessible from Anywhere:

Caregivers or family members can open the dashboard on their mobile, laptop, or tablet from any place as long as they are connected to the same network.

Supports Multiple Users:

More than one person can view the dashboard at the same time, making it useful for group monitoring and quick decision-making.

Easy Integration with IoT Devices:

The dashboard connects smoothly with the ESP32 and can be expanded later with cloud storage, notifications, or additional sensors.

User-Friendly Interface:

The dashboard is simple to view and understand, making it easy for caregivers to check the system status without any technical skills.

Low Maintenance:

Since it runs over Wi-Fi, the system does not need frequent service or manual checks, making it reliable for long-term use.

CONNECTIONS:

1.MPU6050 -> ESP32 (I2C)

VCC -> 3.3V (ESP32)

GND -> GND (ESP32)

SDA -> GPIO 21

SCL -> GPIO 22

INT -> GPIO 19

2. Buzzer -> ESP32

+ -> GPIO 18

- -> GND

3. GPS Module -> ESP32 (UART)

VCC -> 5V (check module specs)

GND -> GND

TX -> GPIO 16 (RX2)

RX -> GPIO 17 (TX2)

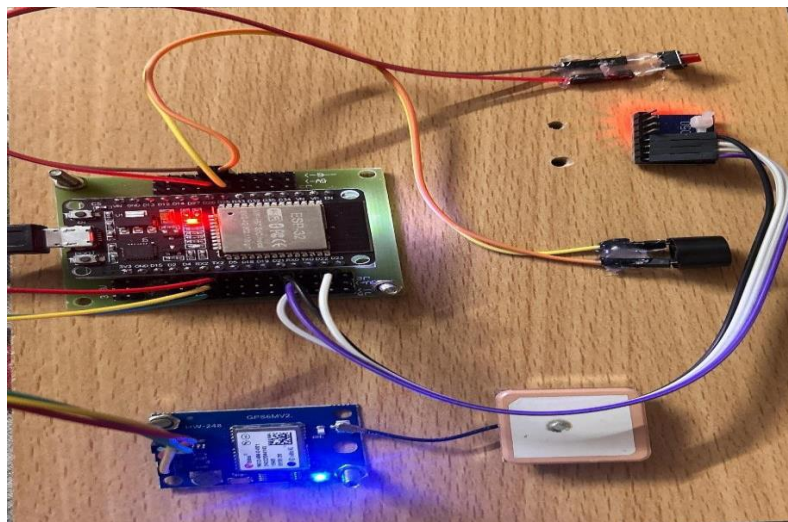


Fig 2:

The photo shows the ESP32 with GPS, buzzer, and sensors connected on a board. It displays the working prototype of the system.

REQUIRED CODE:

```
#include <TinyGPS++.h>

#include <HardwareSerial.h>

#include <WiFi.h>

#include <Wire.h>

#include <Adafruit_MPU6050.h>

#include <Adafruit_Sensor.h>

#include <WebServer.h>

// ----- GPS -----

TinyGPSPlus gps;

HardwareSerial SerialGPS(2); // UART2 for GPS

// ----- WiFi -----

const char* ssid = "SRMAP-TV1";

const char* password = "$rm@p_1TV";

// ----- MPU6050 -----

Adafruit_MPU6050 mpu;

float threshold = 6.0; // Acceleration threshold for fall detection

// ----- BUZZER & BUTTON -----

const int buzzerPin = 25;

const int buttonPin = 26;

bool buzzerActive = false;

bool fallDetected = false;

// ----- GPS Variables -----

double flat = 0.0, flon = 0.0;

int sats = 0;

// ----- MEDICINE ALERT -----

unsigned long lastMedicineTime = 0;

bool medicineAlert = false;

unsigned long medicineAlertStart = 0;

const unsigned long alertDuration = 3000;

// ----- WEB SERVER -----
```

```
WebServer server(80);
```

```
// ----- HTML DASHBOARD -----
```

```
String htmlPage() {
```

```
String html = "<!DOCTYPE html><html><head>";
```

```
html += "<title>ESP32 Dashboard</title>";
```

```
html += "<meta http-equiv='refresh' content='2'>";
```

```
// Font + Icons + Styles
```

```
html += "<link rel='stylesheet' href='https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css'>";
```

```
html += "<style>";
```

```
html += "body { margin:0; background:#eef1f5; font-family:Segoe UI, sans-serif; }";
```

```
html += "header { background:linear-gradient(90deg,#1e3c72,#2a5298); padding:25px; color:white;";
```

```
html += "text-align:center; font-size:28px; font-weight:600; letter-spacing:1px;";
```

```
html += "box-shadow:0 3px 10px rgba(0,0,0,0.2);}";
```

```
html += ".container { display:flex; flex-wrap:wrap; justify-content:center; padding:30px; }";
```

```
html += ".card { background:white; width:280px; padding:25px; margin:15px; border-radius:18px;";
```

```
html += "box-shadow:0 5px 18px rgba(0,0,0,0.08); transition:0.3s;}";
```

```
html += ".card:hover { transform:translateY(-6px); box-shadow:0 8px 25px rgba(0,0,0,0.15);}";
```

```
html += ".card-title { font-size:20px; font-weight:600; margin-bottom:10px; color:#333; }";
```

```
html += ".value { font-size:22px; margin-top:8px; font-weight:700; color:#1e3c72; }";
```

```
html += ".icon { font-size:34px; color:#2a5298; margin-bottom:15px; }";
```

```
html += ".alert-card { background:#ffe5e5; border-left:6px solid #c62828; }";
```

```
html += ".alert-card .value { color:#b71c1c; }";
```

```
html += "</style></head><body>";
```

```
// HEADER
```

```
html += "<header>Fall Detection IoT Dashboard</header>";
```

```
html += "<div class='container'>";
```

```
// GPS Cards
```

```
html += "<div class='card'><i class='fa-solid fa-location-crosshairs icon'></i>";
```

```
html += "<div class='card-title'>Latitude</div><div class='value'>" + String(flat, 6) + "</div></div>";
```

```
html += "<div class='card'><i class='fa-solid fa-location-dot icon'></i>";
```

```
html += "<div class='card-title'>Longitude</div><div class='value'>" + String(flon, 6) + "</div></div>";
```

```
// Fall Detection
```

```
html += "<div class='card'><i class='fa-solid fa-person-falling icon'></i>";
```

```
html += "<div class='card-title'>Fall Detected</div>";
```

```
html += "<div class='value'>" + String(fallDetected ? "YES" : "NO") + "</div></div>";
```

```

// Buzzer

html += "<div class='card'><i class='fa-solid fa-bell icon'></i>";

html += "<div class='card-title'>Buzzer Status</div>";

html += "<div class='value'>" + String(buzzerActive ? "ACTIVE" : "OFF") + "</div></div>";

// Medicine Alert

if (medicineAlert) {

    html += "<div class='card alert-card'><i class='fa-solid fa-capsules icon' style='color:#c62828;'></i>";

    html += "<div class='card-title'>Medicine Alert</div>";

    html += "<div class='value'>Take your medicine now!</div></div>";

}

html += "</div></body></html>";

return html;

}

void handleRoot() {

    server.send(200, "text/html", htmlPage());

}

// ----- SETUP -----

void setup() {

    Serial.begin(115200);

    SerialGPS.begin(9600, SERIAL_8N1, 16, 17);

    Serial.println("GPS Started...");

    Serial.print("WiFi Connecting...");

    WiFi.begin(ssid, password);

    while (WiFi.status() != WL_CONNECTED) {

        delay(300);

        Serial.print(".");

    }

    Serial.println("\nWiFi Connected: " + WiFi.localIP().toString());

    mpu.begin();

    Serial.println("MPU6050 OK!");

    pinMode(buzzerPin, OUTPUT);

    pinMode(buttonPin, INPUT_PULLUP);

    server.on("/", handleRoot);

    server.begin();

    lastMedicineTime = millis();

```

```

}

// ----- LOOP -----

void loop() { server.handleClient();

// GPS reading

while (SerialGPS.available() > 0) {

    gps.encode(SerialGPS.read());

}

if (gps.location.isValid()) {

    flat = gps.location.lat();

    flon = gps.location.lng();

    sats = gps.satellites.value(); // still reads but not displayed

}

// MPU reading

sensors_event_t a, g, temp;

mpu.getEvent(&a, &g, &temp);

// fall detection

if (!buzzerActive && (abs(a.acceleration.x) > threshold || abs(a.acceleration.y) > threshold)) {

    Serial.println("Fall detected!");

    buzzerActive = true;

    fallDetected = true;

    unsigned long start = millis();

    while (millis() - start < 10000) {

        digitalWrite(buzzerPin, HIGH);

        delay(500);

        digitalWrite(buzzerPin, LOW);

        delay(500);

        if (digitalRead(buttonPin) == LOW) {

            Serial.println("Buzzer stopped by user");

            buzzerActive = false;

            fallDetected = false;

            break;

        }

        server.handleClient();

    }

    if (buzzerActive) {

```

```
    digitalWrite(buzzerPin, HIGH);

    delay(2000);

    digitalWrite(buzzerPin, LOW);

    buzzerActive = false;
}

}

// medicine alert
if (millis() - lastMedicineTime >= 60000) {

    medicineAlert = true;

    medicineAlertStart = millis();

    lastMedicineTime = millis();

}

if (medicineAlert && (millis() - medicineAlertStart >= 3000)) {

    medicineAlert = false;

}

delay(200);

}
```

RESULTS:

The elderly care monitoring system using the ESP32 was successfully developed and tested. The MPU6050 sensor accurately detected fall events, and the GPS module provided correct location data. The web dashboard displayed real-time information such as latitude, longitude, fall status, and system activity without delay. The alert system worked smoothly, and the data refreshed instantly through Wi-Fi. Overall, the system performed reliably, responded quickly to fall events, and proved effective for continuous monitoring of elderly individuals..

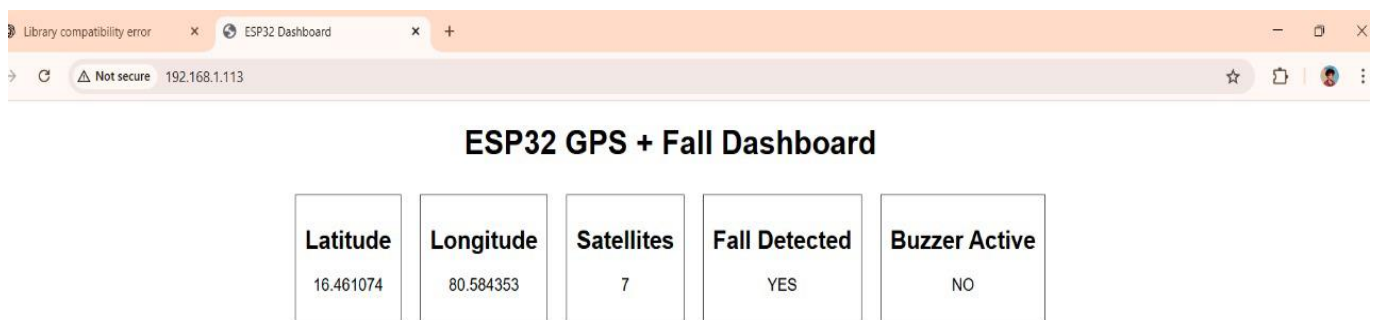


Fig 3:

The dashboard shows real-time GPS data, satellite count, and fall detection status. It also indicates if the buzzer is active. Caregivers can quickly check the user's condition.

CONCLUSION:

The Elderly Care Monitoring System successfully integrates IoT technology to enhance safety and support for senior citizens. By combining the ESP32, MPU6050 sensor, and GPS module, the system provides reliable fall detection and real-time location tracking. The buzzer alert mechanism ensures that emergencies receive immediate attention from nearby caregivers. The push-button feature helps minimize false alarms by allowing the user to cancel the alert if they are safe. The medicine reminder function adds additional value by helping elderly individuals maintain their daily health routine. Through its web-based dashboard, caregivers can remotely monitor the user's status at any time. The continuous data flow and automatic updates improve response time during critical situations. Overall, the system offers a practical and efficient solution for elderly safety and independence. It reduces the risks associated with living alone and increases confidence for both users and caregivers. The project demonstrates how IoT can improve healthcare monitoring in a simple and cost-effective way. This system can be further enhanced in the future by adding cloud storage, mobile alerts, and advanced health sensors.